

NOBEL PRIZE DATASET

Exploratory Data Analysis & Insight Discovery

01 • ASSIGNMENT OBJECTIVE

In the previous assignment, you worked with the Nobel Prize dataset to become familiar with **Pandas**, including data loading, filtering, grouping, aggregation, and basic data manipulation.

In this assignment, the goal is different.

You are now expected to think like a **Data Analyst**.

Instead of being given a list of questions to answer, you will be given a dataset and asked to **discover interesting patterns, relationships, trends, and anomalies on your own**.

Your task is not simply to analyze the data. Your task is to ask good questions about the data and use it to find meaningful answers.

You should explore the dataset from different perspectives and try to discover insights that are not immediately obvious.

02 • MAIN REQUIREMENT

You must discover and explain **at least 12 different insights** from the Nobel Prize dataset.

There is **no fixed list of questions** that you have to answer.

You are encouraged to investigate different dimensions of the dataset, such as:

- Gender
- Age
- Country
- Category
- Year
- Decade
- Time periods
- Nationality
- Multiple winners
- Organizations
- Affiliations
- Geographic distribution
- Relationships between different variables
- Changes over time
- Unusual or exceptional cases

These are only examples of possible directions.

Do not limit your analysis to these dimensions.

A strong analysis should demonstrate your ability to discover questions that were **not explicitly given to you**.

03 • THINK LIKE A DATA ANALYST

For every analysis, try to follow this general thought process:

Step 1 Ask a Question

Start with something you are curious about.

For example:

“Are Nobel Prize winners predominantly male?”

Step 2 Investigate the Question

Use Pandas to manipulate and analyze the data.

You may need to:

- Filter rows
- Group data
- Aggregate values
- Sort results
- Calculate percentages
- Compare groups
- Create new features
- Analyze data over time

Step 3 Visualize the Finding

Whenever visualization can communicate the result better, create an appropriate visualization.

Do not automatically use a bar chart for everything.

Think about **which visualization best represents the question**.

Step 4 Interpret the Result

Do not stop at:

“Male winners = 90%, Female winners = 10%.”

Explain what this means.

For example:

“The dataset shows a strong gender imbalance among Nobel Prize winners. However, this overall pattern may hide differences between Nobel categories. Therefore, the next question is whether the gender distribution is similar across all categories.”

This leads naturally to deeper analysis.

Step 5 Ask a Follow-up Question

Good data analysis is often iterative.

One insight should make you ask another question.

For example:

Overall gender distribution

↓

“Are males still dominant within every Nobel category?”

↓

“Has the gender distribution changed over time?”

↓

“Which category has the largest proportion of female winners?”

↓

“Has this changed across different decades?”

This is the kind of analytical thinking expected in this assignment.

04 • EXAMPLES OF EXPECTED THINKING

The following examples are provided to demonstrate the **style of thinking expected**.

They are **not questions that you must answer**.

Example 1 Hierarchical Analysis

You might begin by asking:

Are Nobel Prize winners predominantly male or female?

You could first investigate the overall distribution.

Then, instead of stopping there, ask:

Does the same pattern exist within every Nobel Prize category?

You could then investigate whether the pattern changes when looking at different time periods.

For example:

Overall

→ Gender distribution

Then

→ Gender distribution by category

Then

→ Gender distribution by decade

Then

→ Category + decade + gender

The important part is not the specific result.

The important part is that you **progressively investigate the data at different levels of detail**.

Example 2 Geographic Analysis

You might ask:

Which countries have produced the largest number of Nobel Prize winners?

A simple approach would be to create a count plot of winners by country.

However, consider whether a **geographical visualization** would communicate this information more effectively.

For example, you could investigate the geographic distribution of Nobel Prize winners and represent it using a **world map with a heatmap/choropleth-style visualization**.

Then ask deeper questions:

- Does the geographic distribution change over time?
- Were particular countries dominant during certain periods?
- Are certain Nobel categories geographically concentrated?
- Did the countries dominating the prizes change between different decades?

Again, these are examples of possible directions, not a predefined checklist.

Example 3 Time-Based Analysis

You might notice that the dataset contains the year in which each prize was awarded.

Instead of simply plotting the number of prizes per year, think about what else can be learned from time.

For example:

Were there periods in which a particular country became unusually dominant?

You could investigate:

Year

→ **Decade**

→ **Country**

→ **Category**

and look for periods where particular countries, categories, or groups of people became unusually prominent.

You could then visualize the result using an appropriate time-series, stacked, heatmap, or other visualization.

05 • GO BEYOND SIMPLE QUESTIONS

Avoid analyses that only answer simple questions such as:

- How many winners are there?
- Which category has the most winners?
- Which country has the most winners?
- How many male and female winners are there?

These can be useful as **starting points**, but they should not be the final insight.

Try to take your analysis one step further.

For example:

Basic analysis

Which category has the most Nobel Prize winners?

Better analysis

How has the number of winners in each category changed over time?

Even deeper analysis

Which categories have experienced the largest changes in representation across different decades, and when did those changes occur?

The goal is to move from “**What?**” toward:

“**Why might this be happening?**”

and

“**What else can I discover from this pattern?**”

You are not necessarily expected to prove causation. However, you should distinguish between **what the data demonstrates** and **your possible explanation or hypothesis**.

06 • VISUALIZATION REQUIREMENTS

Visualization is an important part of this assignment.

You should use visualizations whenever they help communicate your findings.

However, **do not use the same plot for every question**.

Choose the visualization based on the analytical question.

Possible visualizations include:

- Bar charts
- Horizontal bar charts
- Histograms
- Box plots
- Violin plots
- Line charts
- Area charts
- Stacked bar charts
- Heatmaps
- Scatter plots
- Bubble charts
- Pie/donut charts where appropriate
- Geographic maps
- Other appropriate visualizations

For example:

Analytical Question	Possible Visualization
Distribution of ages	Histogram / KDE
Compare categories	Bar chart
Change over time	Line chart
Compare multiple categories over time	Stacked/line chart
Relationship between variables	Scatter plot
Distribution across groups	Box/violin plot
Country distribution	Geographic map
Country × decade	Heatmap
Category × gender	Heatmap / grouped visualization

These are suggestions, **not requirements**.

You should decide which visualization best communicates your finding.

07 • INSIGHT QUALITY MATTERS

Your 12 insights should not be 12 different versions of the same analysis.

For example, these would provide very little analytical diversity:

1. Most common country
2. Second most common country

3. Third most common country
4. Fourth most common country

Instead, try to investigate **different aspects of the dataset**.

Your insights could involve different combinations of:

Who?

→ Gender, age, nationality, organization, etc.

Where?

→ Country, region, geographic distribution, etc.

When?

→ Year, decade, historical period, etc.

What?

→ Nobel category, prize type, affiliation, etc.

How did it change?

→ Trends over time

How are variables related?

→ Category × gender, country × category, age × category, etc.

Are there exceptions?

→ Unusual winners, unusual years, unusual countries, unexpected patterns, etc.

The strongest analyses will often combine **multiple dimensions**.

08 • YOUR 12 INSIGHTS

For each insight, include the following:

Insight #1

Question:What question are you investigating?

Analysis:Explain briefly how you analyzed the data.

Visualization:Include an appropriate visualization.

Finding:What did you discover?

Interpretation:Why is this finding interesting or important?

Follow-up:If applicable, mention another question that your finding raised.

Repeat this structure for at least **12 different insights**.

09 • DON'T JUST REPORT NUMBERS

A Data Analyst does not simply produce tables of numbers.

For example:

“USA: 400 winnersUK: 200 winnersGermany: 180 winners”

This is a result, but by itself it is not a strong insight.

Instead, ask:

Why does this distribution look this way?

Then investigate further.

For example, you might discover that the distribution changes substantially when looking at different decades or Nobel categories.

Your goal is to turn **data → analysis → visualization → insight**.

10 • COMPARE BEFORE YOU CONCLUDE

Be careful when making general statements.

For example:

“Country A dominates Nobel Prizes.”

Before making this conclusion, consider:

- Does Country A dominate every category?
- Does it dominate every decade?
- Is the pattern driven by one particular category?
- Does the pattern change over time?
- Are there missing values?
- Are you counting people or prizes?
- Can multiple people share the same prize?
- Is the country information representing birthplace, citizenship, or affiliation?

A good analyst always asks:

“Is my conclusion actually supported by the way the data is structured?”

11 • DATA QUALITY MATTERS

Before drawing conclusions, inspect the dataset carefully.

Look for:

- Missing values
- Duplicate records
- Inconsistent categories
- Inconsistent country names
- Unexpected values
- Changes in how information was recorded
- Multiple winners for the same prize
- Winners associated with multiple organizations/countries
- Other potential data-quality issues

If you encounter a limitation in the dataset that could affect your analysis, **mention it explicitly**.

12 • CREATIVITY IS ENCOURAGED

There is no single “correct” set of 12 insights.

Two students may analyze the same dataset and produce completely different analyses.

That is expected.

In fact, discovering an interesting relationship that nobody else found is a **good sign**.

You are encouraged to explore questions that are not mentioned in these guidelines.

Think beyond:

“What questions was I told to answer?”

Instead ask:

“What can I discover that I was not told to look for?”

13 • FINAL DELIVERABLE

Submit a Jupyter Notebook containing:

A. Data Understanding

- Load the dataset
- Describe the dataset
- Explain the important columns
- Check data quality

B. Exploratory Analysis

- Perform appropriate preprocessing where necessary
- Explore relevant variables
- Investigate relationships between variables

C. Insight Discovery

- At least **12 meaningful and different insights**
- Each insight should include:
 - Question
 - Analysis
 - Visualization where appropriate
 - Finding
 - Interpretation

D. Final Summary

At the end of your notebook, provide a section titled:

Key Findings

Summarize your **most important 5–7 discoveries** from the analysis.

You should also answer:

What surprised you most about the Nobel Prize dataset?

and

What additional analysis would you perform if you had more information or additional data?

14 • ASSESSMENT CRITERIA

Your grade will not depend only on whether your code runs correctly.

You will be evaluated on your ability to **think analytically**.

Criterion	Description
Data Understanding	Understanding the structure and meaning of the dataset
Pandas Skills	Effective use of filtering, grouping, aggregation, transformation, etc.
Insight Quality	How meaningful and interesting your discoveries are
Analytical Depth	Going beyond simple counts and rankings
Visualization	Choosing appropriate visualizations to communicate findings
Interpretation	Clearly explaining what the results mean
Diversity	Exploring different dimensions and relationships
Data Quality	Identifying and appropriately handling data issues
Creativity	Discovering non-obvious patterns
Communication	Presenting the analysis clearly and logically

★ FINAL CHALLENGE

Imagine that you are given this dataset by a company, research group, or organization and **nobody tells you what questions to ask.**

Your job is to investigate the data and find something interesting.

There are no predefined answers.

There are no predefined 12 questions.

There is no requirement for everyone to discover the same insights.

Your objective is to demonstrate that you can look at a dataset, ask meaningful questions, investigate them, visualize the evidence, and communicate what you discovered.

Don't just analyze the data. Interrogate it.